

[7]
✓ 30s

```
from google.colab import drive
drive.mount('/content/drive')

import os

test_dir = "/content/drive/MyDrive/test_data"

for category in os.listdir(test_dir):
    folder_path = os.path.join(test_dir, category)
    if os.path.isdir(folder_path):
        num_images = len(os.listdir(folder_path))
        print(f"{category}: {num_images} images")
```

▼ ... Mounted at /content/drive
hospital: 19 images
classroom: 17 images
roadside: 16 images

[8]
✓ 1m

```
import glob
from PIL import Image
import torch

true_labels = []
predicted_labels = []

def run_pipeline(image_path):
    # Object detection
    results = yolo_model(image_path, verbose=False)
```

8]

```
true_labels = []
predicted_labels = []

def run_pipeline(image_path):

    results = yolo_model(image_path, verbose=False)
    result = results[0]
    detected_objects = [
        (yolo_model.names[int(box.cls[0])], float(box.conf[0]))
        for box in result.bboxes
    ]

    img = Image.open(image_path).convert('RGB')
    input_tensor = preprocess(img).unsqueeze(0)
    with torch.no_grad():
        probs = torch.nn.functional.softmax(scene_model(input_tensor), dim=1)[0]
    top5_prob, top5_idx = probs.topk(5)
    scene_predictions = [(classes[idx], float(prob)) for prob, idx in zip(top5_prob, top5_idx)]

    scores = compute_category_scores(detected_objects, scene_predictions, category_map)
    final_category = max(scores, key=scores.get) if max(scores.values()) > 0 else "uncategorized"
    return final_category

for category_folder in os.listdir(test_dir):
    folder_path = os.path.join(test_dir, category_folder)
    if not os.path.isdir(folder_path):
        continue
```

```

        return final_category

    for category_folder in os.listdir(test_dir):
        folder_path = os.path.join(test_dir, category_folder)
        if not os.path.isdir(folder_path):
            continue
        for image_path in glob.glob(os.path.join(folder_path, "*")):
            try:
                predicted = run_pipeline(image_path)
                true_labels.append(category_folder)
                predicted_labels.append(predicted)
            except Exception as e:
                print(f"Skipped {image_path}: {e}")

    print(f"\nProcessed {len(true_labels)} images total")

```

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Processed 52 images total

```

[9] 2s from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
import matplotlib.pyplot as plt
import seaborn as sns

accuracy = accuracy_score(true_labels, predicted_labels)
print(f"Overall accuracy: {accuracy:.3f}\n")

print("Classification report:")
print(classification_report(true_labels, predicted_labels))

labels_sorted = sorted(set(true_labels) | set(predicted_labels))
cm = confusion_matrix(true_labels, predicted_labels, labels=labels_sorted)

plt.figure(figsize=(8, 6))
sns.heatmap(cm, annot=True, fmt='d', xticklabels=labels_sorted, yticklabels=labels_sorted, cmap='Blues')
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.title('Confusion Matrix')
plt.tight_layout()
plt.show()

```

...

Classification report:				
	precision	recall	f1-score	support
classroom	0.81	1.00	0.89	17
hospital	0.75	0.79	0.77	19
roadside	0.90	0.56	0.69	16
uncategorized	0.00	0.00	0.00	0
accuracy			0.79	52
macro avg	0.61	0.59	0.59	52
weighted avg	0.82	0.79	0.79	52

